RAK5802 WisBlock RS485 Interface Module Datasheet

Overview

Description

The RAK5802 WisBlock Interface module was designed to be part of a production-ready IoT solution in a modular way and must be combined with a WisBlock Core and a Base module.

The RAK5802 is an RS485 Modbus extension module that allows you to provide IoT connectivity to a new/existing solution with an RS485 interface. This module converts the RS485 signals into UART signals. These signals are collected by an MCU located inside of a WisBlock Core module for further data transmission.

The RAK5802 module features one RS485 interface. Inside, a protection circuity is added to protect against ESD hazards. The RAK5802 module complies with the IEC61000-4-2 standard and can protect up to 18 kV HBM ESD.

In addition, the RAK5802 supports one I2C interface, customers can use this I2C interface to connect external sensors.

Features

- RS485 interface and Modbus protocol
- Compatible with multiple WisBlock Core modules, such as RAK4631.
- Supports up to 16 RS485 nodes
- Supports TX/RX signals automatic detection, no need to have DE signal for the RS485 chip.
- Low power consumption mode. The module can be powered off by the WisBlock Core module for saving energy during idle periods.
- Reserved I2C expansion interface
- In the field fast crimping terminals
- Designed with 18 kV ESD protection level
- Chipset: 3PEAK TP8485E
- Small dimensions of 35 mm x 25 mm

Specifications

Overview

Block Diagram

Figure 1 shows the block diagram of the RAK5802 module. Internally, the RS485 converter translates the RS422 electrical differential signal into the UART signal. This component can be disabled by the Enable signal through the MCU in order to save power during idle periods.

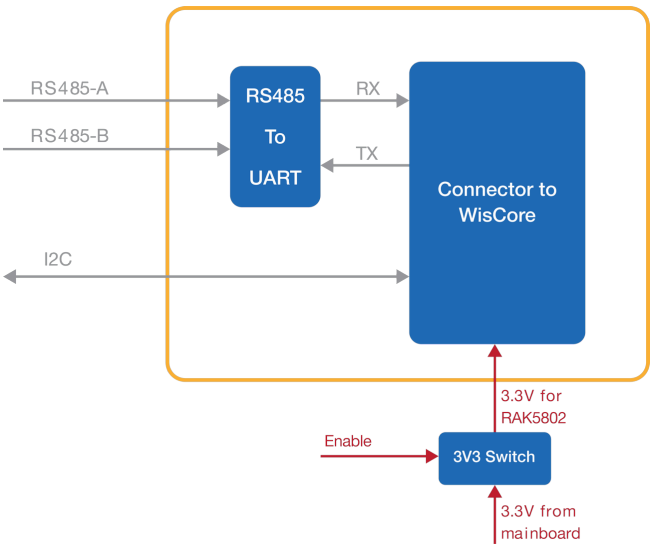


Figure 1: RAK5802 block diagram

Hardware

The hardware specification is categorized into four parts. It discusses the pinouts of the module and its corresponding functions and diagrams. It also covers the electrical and mechanical characteristics that include the tabular data of the functionalities and standard values of the RAK5802 WisBlock Module.

Chipset

Vendor	Part number
3PEAK	TP8485E

Pin Definition

Figure 2 shows the pin number of the sensor connector of the RAK5802 module.

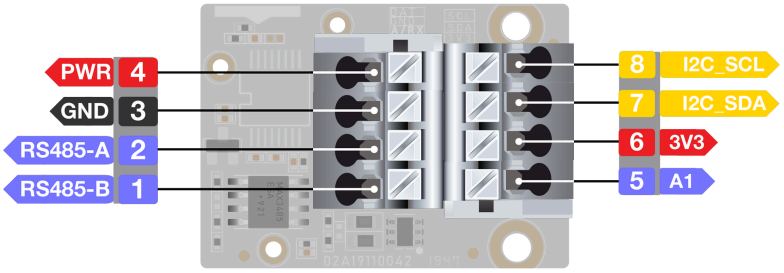


Figure 2: RAK5802 sensor connector

The pin definition of the sensor connector is described in the table below:

Pin number	Description	Function Description
1	RS485-B	RS485-B
2	RS485-A	RS485-A
3	GND	GND
4	PWR	Power supply from battery on mainboard

Pin number	Description	Function Description
5	A1	Analog 1 to MCU
6	3V3	3V3 input
7	I2C_SDA	SDA for I2C interface
8	I2C_SCL	SCL for I2C interface

WisBlock Interface Connector

Figure 3 shows the pin order of the IO connector on the module. Through this connector, the RAK5802 module is attached to the WisBoard baseboard.

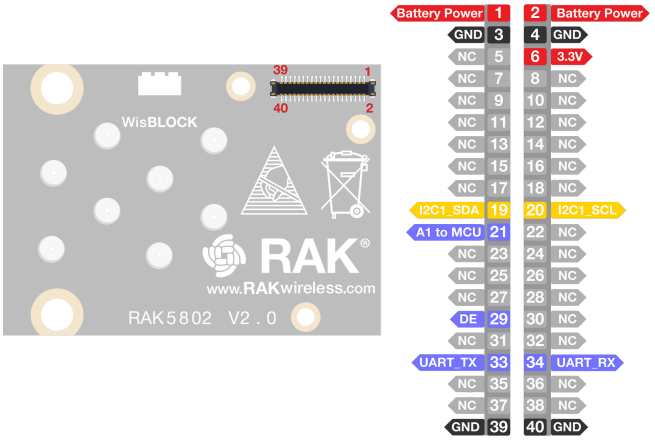


Figure 3: RAK5802 internal WisBlock Interface connector

The table below shows the functionalities of each pin of the WisBlock Interface connector:

Pin number	Description	Pin number	Description
1	Battery power	2	Battery power
3	GND	4	GND
5	NC, reserved for 3V3	6	3.3 V power
7	NC	8	NC
9	NC	10	NC
11	NC	12	NC
13	NC	14	NC
15	NC	16	NC
17	NC	18	NC
19	SDA for I2C1	20	SCL for I2C1
21	Analog1 to MCU	22	NC
23	NC	24	NC

Pin number	Description	Pin number	Description
25	NC	26	NC
27	NC	28	NC
29	Reserved for DE	30	NC
31	NC	32	NC
33	UART_TX	34	UART_RX
35	NC	36	NC
37	NC	38	NC
39	GND	40	GND

Electrical Characteristics

Absolute Maximum Ratings

Symbol	Description	Min.	Nom.	Max.	Unit
VBAT	Power supply for the module	-0.5		4.2	V
Iout	Step down IC output current			1000	mA
TEMP	Working temperature	-30		65	°C
Storage	Storage Temperature	-40		85	°C

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VBAT	Power supply for the module	2.6		4.2	V
3V3	3.3 V power supply	3.0	3.3	3.6	V

Mechanical Characteristics

Board Dimensions

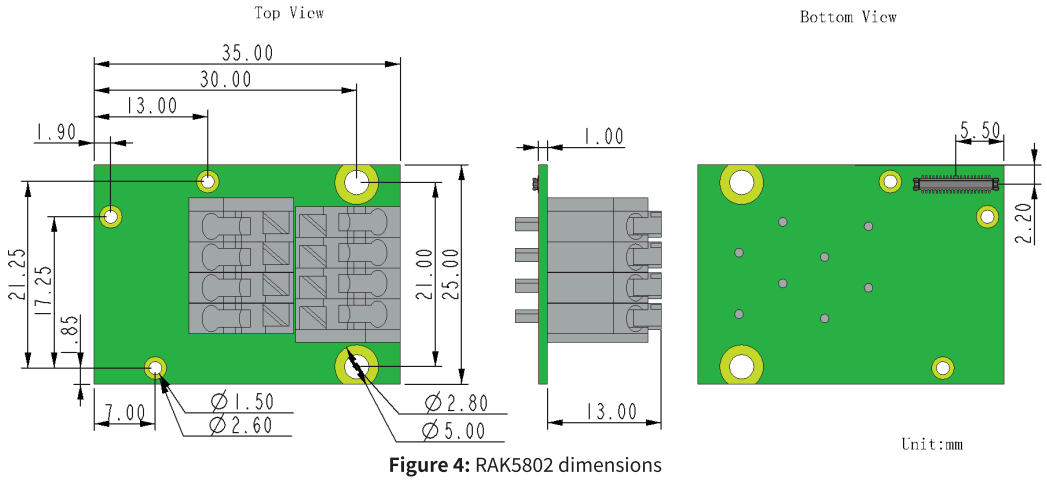
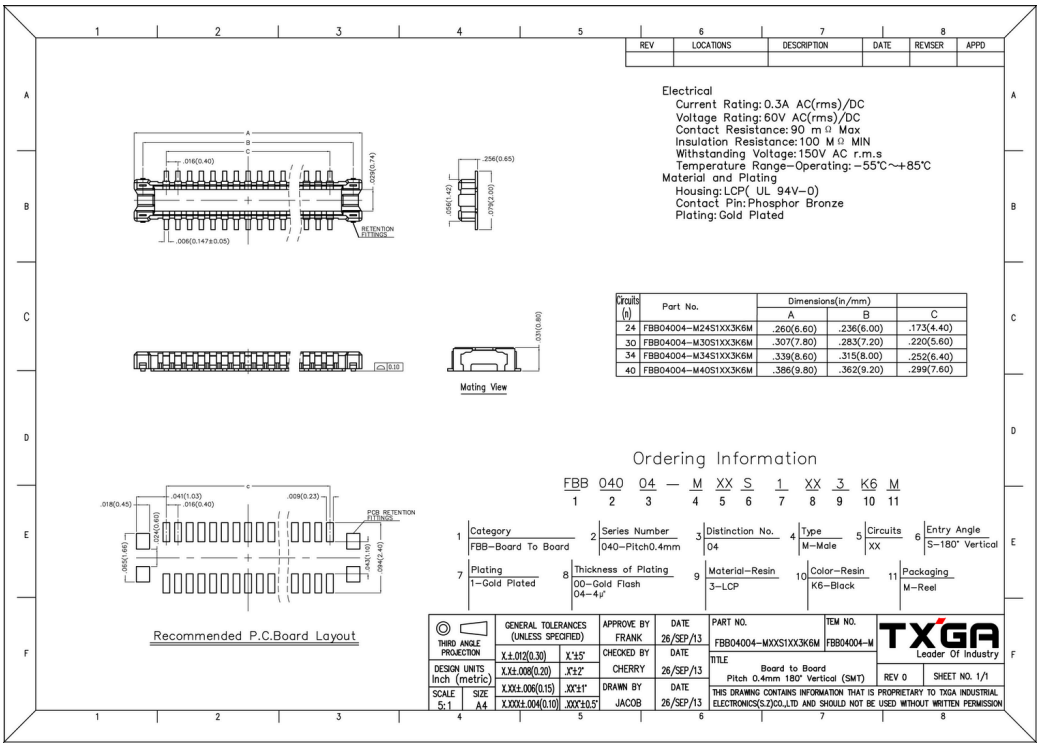


Figure 4: RAK5802 dimensions

WisConnector PCB Layout



Circuits (n)	Part No.	Dimensions(in/mm) A	B	C
24	FBB04004—M24S1XX3K6M	.260(6.60)	.236(6.00)	.173(4.40)
30	FBB04004—M30S1XX3K6M	.307(7.80)	.283(7.20)	.220(5.60)
34	FBB04004—M34S1XX3K6M	.339(8.60)	.315(8.00)	.252(6.40)
40	FBB04004—M40S1XX3K6M	.386(9.80)	.362(9.20)	.299(7.60)

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

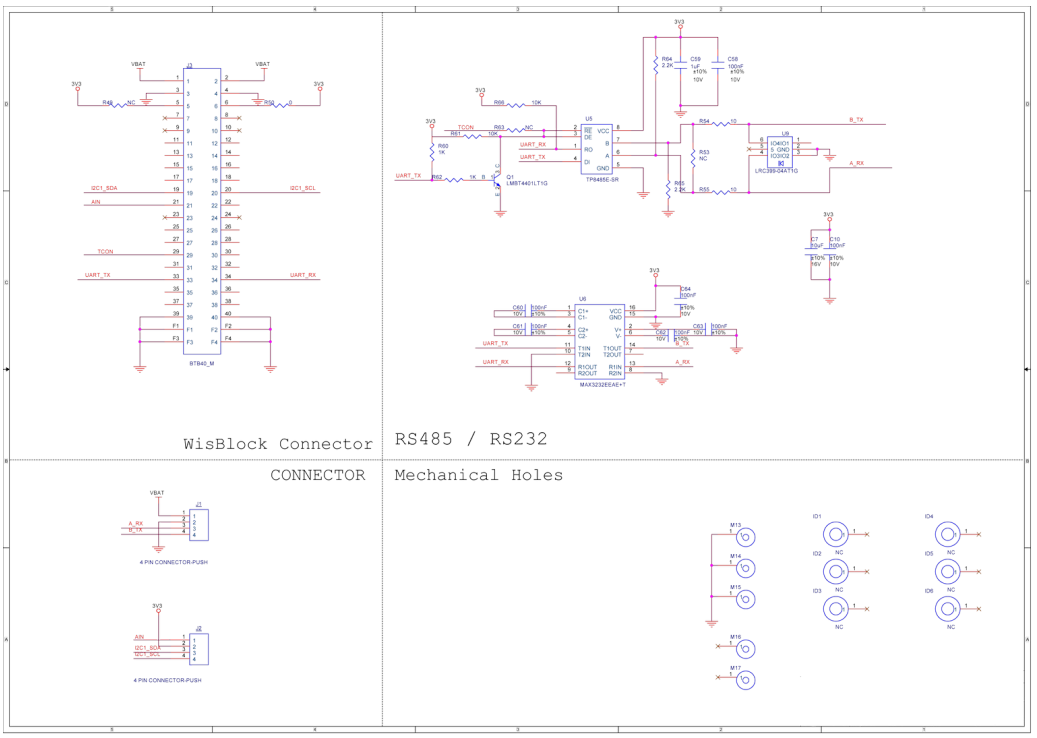


Figure 6: RAK5802 schematic diagram

RAK5802 and WisBlock Compatibility

In the ecosystem of WisBlock, the philosophy is to allow you to combine a WisBlock module of one category with a WisBlock module of other categories. This is true in most cases, but it is important to review the compatibility matrix when designing a customized solution.

	Compatible Wisblock Module	Description
WisBlock Base board	RAK5005/RAK5005-O	RAK5802 can be attached in the IO slot of the RAK5005-O base board.
WisBlock Core module	RAK4631	RAK5802 is compatible with RAK4631 WisBlock Core module.
	RAK4201	Please select RAK4201L-485 for low band or RAK4201H-485 for high band.
	RAK4202	RAK5802 is compatible with RAK4202 WisBlock Core module.
	RAK4261	RAK5802 is compatible with RAK4261 WisBlock Core module.
	RAK4601	RAK5802 is compatible with RAK4601 WisBlock Core module.